REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

A Particle Model of Waves Practice Questions

Grade 12 Advanced | Module 22 Lesson 1 | A Particle Model of Waves

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: $E_{photon} = hf = hc/\lambda$; $KE_{max} = hf - \varphi$; $f_0 = \varphi/h$; $p_{photon} = h/\lambda$. $h = 6.626 \times 10^{-34}$ J s; $c = 3.00 \times 10^{8}$ m/s; 1 eV = 1.602×10^{-19} J.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 12 book, printed pages 46–58.

P01. A photon is:

A. An electron inside a metal B. A quantum of electromagnetic radiation
C. A classical orbit of an atom D. A neutron with charge

P02. Below a metal's threshold frequency, increasing light intensity alone causes:

A. Greater maximum electron kinetic energy B. Emission at every possible intensity
C. No photoelectron emission in the ideal single-photon model D. The threshold frequency to become zero

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P03. Above threshold, increasing frequency with intensity otherwise controlled increases:

A. Electron rest mass B. Metal atomic number
C. Planck's constant D. Maximum photoelectron kinetic energy

P04. The slope of a KE_{max} versus frequency graph is:

A. Planck's constant h B. The work function directly
C. The speed of light D. The electron charge in coulombs only

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P05. The Compton effect supports the idea that photons possess:

A. Electron rest mass B. Momentum
C. A fixed electric charge D. Zero energy

P06. Find the energy of a photon of frequency 2×10^{14} Hz. Use $h = 6.626 \times 10^{-34}$ J s.

A. 2.65×10^{-19} J B. 6.63×10^{-20} J
C. 1.33×10^{-19} J D. -1.33×10^{-19} J

P07. Find the energy of a photon of wavelength 500 nm. Use $h = 6.626 \times 10^{-34}$ J s and $c = 3.00 \times 10^{8}$ m/s.

A. 7.95×10^{-19} J B. 1.99×10^{-19} J
C. -3.98×10^{-19} J D. 3.98×10^{-19} J

P08. A photon has energy 4 eV and the metal work function is 2.0 eV. Find the maximum photoelectron kinetic energy.

A. 2 eV B. 4 eV
C. 1 eV D. -2 eV

P09. A metal work function is 3.84×10^{-19} J. Find its threshold frequency. Use h = 6.626×10^{-34} J s.

A. 1.16×10^{15} Hz B. 5.8×10^{14} Hz
C. 2.9×10^{14} Hz D. -5.8×10^{14} Hz

P10. Calculate the momentum of a photon with wavelength 500 nm. Use h = 6.626×10^{-34} J s.

A. 2.65×10^{-27} kg m/s B. 6.63×10^{-28} kg m/s
C. 1.33×10^{-27} kg m/s D. -1.33×10^{-27} kg m/s

P11. Define the photoelectric effect.

P12. Define work function.

P13. Define threshold frequency.

P14. Explain how light intensity affects photoelectric current above threshold.

P15. What is an electron volt?

P16. Find the energy of a photon of frequency 3×10^{14} Hz. Use h = 6.626×10^{-34} J s.

P17. Find the energy of a photon of frequency 4×10^{14} Hz. Use h = 6.626×10^{-34} J s.

P18. Find the energy of a photon of wavelength 550 nm. Use $h = 6.626 \times 10^{-34}$ J s and $c = 3.00 \times 10^{8}$ m/s.

P19. Find the energy of a photon of wavelength 600 nm. Use $h = 6.626 \times 10^{-34}$ J s and $c = 3.00 \times 10^{8}$ m/s.

P20. A photon has energy 4.5 eV and the metal work function is 2.0 eV. Find the maximum photoelectron kinetic energy.

P21. A photon has energy 5 eV and the metal work function is 2.0 eV. Find the maximum photoelectron kinetic energy.

P22. A metal work function is 4.17×10^{-19} J. Find its threshold frequency. Use $h = 6.626 \times 10^{-34}$ J s.

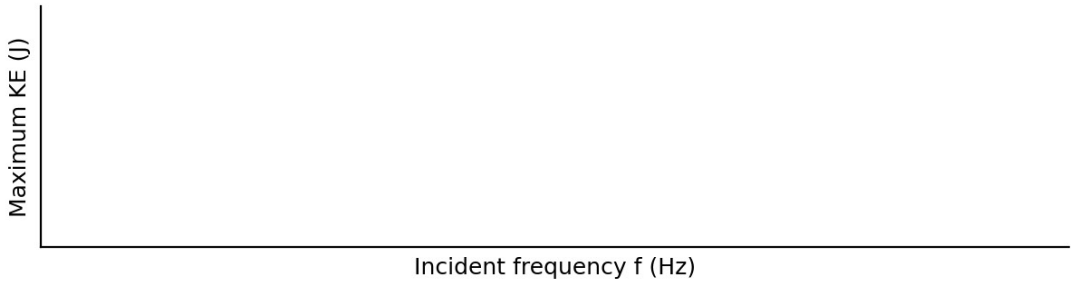
P23. A metal work function is 4.49×10^{-19} J. Find its threshold frequency. Use $h = 6.626 \times 10^{-34}$ J s.

P24. Calculate the momentum of a photon with wavelength 550 nm. Use $h = 6.626 \times 10^{-34}$ J s.

P25. Calculate the momentum of a photon with wavelength 600 nm. Use h = 6.626 × 10^{-34} J s.

P26. A metal has work function 2.0 eV and is illuminated by 3.5 eV photons. Find KEmax and explain the effect of doubling intensity.

P27. Sketch a KEmax–frequency graph for a metal. Label threshold frequency and explain slope.



P28. Why did immediate emission above threshold challenge a purely classical energy-accumulation explanation?

P29. Describe how a graph can be used to estimate Planck's constant.

P30. Explain the wavelength increase in a Compton-scattered photon qualitatively.